

Multiple Choice (10 marks)

1. Which of the following is true of interrupts?
 - (a) They are generated when memory cycles are "stolen".
 - (b) They are used in place of data channels.
 - (c) They can indicate completion of an I/O operation.
 - (d) They cannot be generated by arithmetic operations.
2. Which of the following is NOT a property of bitmap graphics?
 - (a) Fast hardware exists to move blocks of pixels efficiently.
 - (b) Realistic lighting and shading can be done.
 - (c) All line segments can be displayed as straight.
 - (d) Polygons can be filled with solid colors and textures.
3. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
4. Of the following problems concerning a given undirected graph G, which is currently known to be solvable in polynomial time?
 - (a) Finding a longest simple cycle in G
 - (b) Finding a shortest cycle in G
 - (c) Finding ALL spanning trees of G
 - (d) Finding a largest clique in G
5. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
 - (a) Insertion sort
 - (b) Quicksort
 - (c) Merge sort
 - (d) Selection sort

True / False (10 marks)

Answer True or False

6. True or False: Having several different classes of registers is not generally considered an obstacle to aggressive pipelining of an integer unit.
7. True or False: A 3-way, set-associative cache is only effective when 3 or fewer processes are running on the processor.
8. True or False: The language $\{ww \mid w \text{ in } (0 + 1)^*\}$ is context-free and can be accepted by a pushdown automaton.
9. True or False: Identifier length is best specified using a context-free grammar in programming languages.
10. True or False: Python requires manual memory management and cannot be integrated with other programming languages.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find nth number in a sequence which is not a multiple of a given number.
12. Write a function to remove the parenthesis area in a string.

End of Paper